MIL-STD-106A
22 May 1972
SUPERSEDING
MIL-STD-106
16 August 1951

MILITARY STANDARD

MATHEMATICAL SIGNS AND SYMBOLS

FOR USE IN

PHYSICAL SCIENCES AND TECHNOLOGY

FSC DRPR

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T21:08Z
Check the source to verify that this is the current version before use.

DEPARTMENT OF DEFENSE
Washington, D.C. 20301

Mathematical Signs and Symbols for Use in Physical Sciences and Technology.

MIL-STD-106A

1. This Military Standard is approved 22 May 1972, and is mandatory for use by all Departments and Agencies of the Department of Defense.

2. Recommended corrections, additions, or deletions should be addressed to Commander, Naval Ship Engineering Center, Center Building, Prince George's Center, Hyattsville, Maryland 20782.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T21:08Z
Check the source to verify that this is the current version before use.

CONTENTS

SECTION		PAGE
1.	SCOPE	1
1.2	Presentation	1
1.3	ISO Representation	1
2.	REFERENCED DOCUMENTS	1
2.1	Standards and Other Documents	1
3.	DEFINITIONS (Not applicable)	1
4.	GENERAL REQUIREMENTS	1
4.1	Conventions	1
4.2	Mathematical Signs and Symbols Used In Existing Technical Documents	1
4.3	Application	1
5.	DETAILED REQUIREMENTS	1
5.1	Mathematical Signs and Symbols	1
5.2	Arithmetic and Algebra	2
.3	Elementary Geometry	3
5.4	Analytic Geometry	3
5.5	Trigonometry	4
5.6	Hyperbolic Functions	4
5.7	Calculus	5
5.8	Special Functions	5
5.9	Vector Analysis	6
6.	NOTES	6
6.1	International Standardization Agreement	6
APPENDIX		
10.1	Revised or Deleted Symbols	7
INDEX		
	Alphabetical	9

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T21:08Z
Check the source to verify that this is the current version before use.

1. SCOPE

1.1 The mathematical signs and symbols in this standard are for use in arithmetic and algebra, elementary geometry, analytic geometry, trigonometry, hyperbolic functions, calculus, vector analysis, and special functions.

1.2 *Presentation.* The material contained herein is grouped so that symbols having similar or converse meanings are grouped together for convenience.

1.3 *ISO representation.* Certain signs and symbols included herein, and identified by the symbol ISO, are recommended by the International Organization for Standardization (ISO) as standard within the international field. These have been identified within this document for information purposes only, and does not constitute an endorsement or acceptance by the Department of Defense of the ISO Recommendation R31, Part XI.

2. REFERENCED DOCUMENTS

2.1 The latest DoD approved issues of the following documents in effect on the date of invitation for bids form a part of this standard to the extent specified herein.

GOVERNMENTAL

MIL-STD-12, Abbreviations for Use on Drawings, Specifications, Standards, and in Technical Documents

NONGOVERNMENTAL

Quantity Symbols:

ANSI Y10.3-1968, Letter Symbols for Quantities Used in Mechanics of Solid Bodies

ANSI Y10.5-1968, Letter Symbols for Quantities Used in Electrical Science and Electrical Engineering

or

IEEE Std No. 280, Letter Symbols for Quantities Used in Electrical Science and Electrical Engineering

Unit Symbols:

ANSI Y10.19-1969, Letter Symbols for Units Used in Science and Technology

or

IEEE Std No. 260, Letter Symbols for Units Used in Science and Technology

(American National Standards Institute
1430 Broadway
New York, N.Y. 10018)

(The Institute of Electrical and Electronics Engineers, Inc.
United Engineering Building
345 E. 47th St.
New York, N.Y. 10017)

Chemical Symbols:

International Union of Pure and Applied Chemistry Manual, Symbols and Terminology for Physiochemical Quantities and Units.

(Division of Chemistry and Chemical Technology
National Research Council
Washington, D.C. 20418)

3. DEFINITIONS. (Not applicable)

4. GENERAL REQUIREMENTS

4.1 *Conventions.* By convention, the first few lower case letters of the English alphabet ($a, b, c, \ldots$) are generally used to denote constant terms or coefficients and the last few letters of the English alphabet ($\ldots x, y, z$) are generally used to indicate variables. Greek letters usually indicate specific constants except α, β, θ, and ϕ are commonly used to designate angles. To simplify complicated expressions containing numerous or often repeated terms, the substitution of a single capital English letter for a single factor is recommended; thus, the term ($b^2 - 4ac$) may be replaced by D where $D=b^2-4ac$.

4.2 *Mathematical signs and symbols used in existing technical documents.* Unless otherwise specified, engineering drawings, and so forth, using the signs and symbols according to the previous edition (MIL-STD-106), may be revised using the signs and symbols from this edition (MIL-STD-106A).

4.3 *Application.* This standard provides a list of mathematical signs and symbols authorized for use in scientific and engineering publications and studies, and also for use on drawings, specifications, standards, and other technical documents.

NOTE: The following guide will be of use in determining whether signs, symbols, letter symbols, or abbreviations should be used:

(a) Signs, symbols, and letter symbols that appear in this standard and in accordance with ANSI Y10.3, ANSI Y10.5, ANSI Y10.19 and the IUPAC Manual for Symbols and Terminology are to be used in formulas and equations, for example, I=5A or p=3 lbf/in^2

(b) Abbreviations according to MIL-STD-12 are generally used in text (general and local notes on drawings) and are not used in formulas and equations, for example, the current is 5 amp; and the pressure is 3 psi

(c) Do not intermingle abbreviations and symbols in the same expression

5. DETAILED REQUIREMENTS

5.1 MATHEMATICAL SIGNS AND SYMBOLS

Source: https://assist.dla.mil -- Downloaded: 2026-10-23121:08Z
Check the source to verify that this is the current version before use.

5.2 ARITHMETIC AND ALGEBRA

ISO Ref.	ITEM No.	SIGN or SYMBOL	DESCRIPTION
	5.2.1		(Not used)
ISO	5.2.2	+	Addition; positive value; approach through positive values
ISO	5.2.3	−	Subtraction; negative value; approach through negative values
	5.2.4	±	Add or subtract; plus or minus value
	5.2.5	∓	Used where ± has appeared previously as in $(a \pm b)(a^2 \mp ab + b^2) = a^3 \pm b^3$ upper signs are to be taken throughout or else lower signs
ISO	5.2.6	· or ×	Multiplication (dot centered)
	5.2.7	( )	Parentheses; for grouping
	5.2.8	[ ]	Brackets; for grouping
	5.2.9	{ }	Braces; for grouping
	5.2.10	$\overline{\quad}$ (overscript)	Vinculum; for grouping. Bar; for repetition, for example, $1/3 = 0.\overline{3}$
	5.2.11	%	Percent, per hundred
ISO	5.2.12	/	Solidus; indicating division (preferred for running text)
	5.2.13	—	Horizontal rule, indicating division; fraction line
	5.2.14	÷	Division sign; used chiefly in arithmetic (should be replaced by solidus when convenient)
	5.2.15	:	Ratio (in proportion)
	5.2.16	∷	Equals (in proportion)
ISO	5.2.17	=	Equivalent sign; is equal to
ISO	5.2.18	≙	Corresponds to
ISO	5.2.19	≠	Is not equal to
ISO	5.2.20	≈	Is approximately equal to
	5.2.21	∼	Is asymptotic to
ISO	5.2.22	≡	Is identical to
	5.2.23	$\underline{\underline{\text{d}}}$ or $\underline{\underline{\text{def}}}$	Equals by definition
ISO	5.2.24	<	Is less than
	5.2.25	<<	Is much less than
ISO	5.2.26	⋘	Is much less than
ISO	5.2.27	≦ or ≤	Is less than or equal to; not greater than
ISO	5.2.28	>	Is greater than
	5.2.29	>>	Is much greater than
ISO	5.2.30	⋙	Is much greater than
ISO	5.2.31	≧ or ≥	Is greater than or equal to; not less than
ISO	5.2.32	∝	Varies directly as
ISO	5.2.33	n !	Factorial number; continued product of all integers from 1 to n, where n is an integer
ISO	5.2.34	a^n (superscript numbers or letters)	Exponent; a raised to power n (exponent indicates number of iterations, when n is an integer)
ISO	5.2.35	$\sqrt[n]{}$	Radical sign; superscript n indicates order of root (omitted in case of square root); usually accompanied by a vinculum (5.2.10), for example, $\sqrt{a^2 + b^2}$
	5.2.36	$^{m/n}$ (superscript)	Fractional index; . . . raised to power of degree m/n
	5.2.37	$^{-n}$ (superscript)	Negative exponent; for example, $a^{-n} = 1/a^n$
	5.2.38	exp $f(x,y, \ldots)$	Functional symbol; exponential function
	5.2.39	exp u	Functional symbol; exponential u
	5.2.40	i or j	Imaginary unit; j operator $\sqrt{-1}$
	5.2.41	$a \cdot 10^n$ or $a \times 10^n$	Scientific notation; notation by powers of 10
	5.2.42	•	Decimal point (placed on line). Separates integral part from fractional part of a number
ISO	5.2.43	∞	Infinity symbol
ISO	5.2.44	→	Arrow; approaches as a limit

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T21:08Z
Check the source to verify that this is the current version before use.

ISO Ref.	ITEM No.	SIGN or SYMBOL	DESCRIPTION
	5.2.45	$'$ (superscript)	Prime; notational method of distinguishing between differing variables or constants, using the same base letter, as a and a'
	5.2.46	$''$ (superscript)	Double prime; notational method of distinguishing between differing variables or constants, using the same base letter, as a and a''
	5.2.47	$'''$ (superscript)	Triple prime notational method of distinguishing between differing variables or constants, using the same base letter, as a and a'''
	5.2.48	. . .	Three dots; dots of omission, meaning "and so forth"
ISO	5.2.49	$\log_a x$	Logarithm of x to base a
	5.2.50	$\log x$	Logarithm of x to base 10. (common system of logarithms)
ISO	5.2.51	$\ln x$ or $\log_e x$	Logarithm of x to base e (Naperian system or Natural Logarithm)
ISO	5.2.52	e	Base of Naperian (natural Logarithms (2.7182 . . .)
	5.2.53	$P(n, r)$	Permutations of n things taken r at a time
	5.2.54	$C(n, r)$	Combinations of n things taken r at a time
ISO	5.2.55	$\lvert\ \rvert$	Vertical bars; indicates absolute value of the quantity inside the bars; vector magnitude; determinant, for example, $\lvert x \rvert$
	5.2.56	$\lVert\ \rVert$	Double bars indicate a matrix or a norm, for example, $\lVert f \rVert$
	5.2.57	a_{ij}	Element in row i, column j of determinant or matrix
	5.2.58	det (a_{ij})	Determinant of matrix (a_{ij})
	5.2.59	space or half space	Used, instead of commas, to separate convenient groups of digits
	5.2.60	subscript number or letter	Notational method of indicating differing values in a set or series

5.3 ELEMENTARY GEOMETRY

ISO Ref.	ITEM No.	SIGN or SYMBOL	DESCRIPTION
	5.3.1	$\angle$, $\angle_s$	Angle(s)
	5.3.2	$\perp$, $\perp_s$	perpendicular to; Perpendicular(s)
	5.3.3	$\parallel$, $\parallel_s$	parallel to; Parallel(s)
	5.3.4	$\triangle$, $\triangle_s$	Triangle(s)
	5.3.5	$\bigcirc$, $\bigcirc_s$	Circle(s)
	5.3.6	$\square\!\!/$, $\square\!\!/_s$	Parallelogram(s)
	5.3.7	$\square$, $\square_s$	Square(s)
	5.3.8	$\square\!\!\backslash$, $\square\!\!\backslash_s$	Trapezoid(s)
	5.3.9	$\cong$	Is congruent to
	5.3.10	$\sim$	Is similar to
	5.3.11	$\measuredangle$	Is equiangular
	5.3.12	$\therefore$	Three dots; hence, therefore
	5.3.13	$\overline{AB}$	Vinculum; chord AB of a circle; length of line segment between A and B
	5.3.14	$\overrightarrow{AB}$	Directed segment A to B
	5.3.15	$\overset{\frown}{AB}$	Arc AB of a circle
	5.3.16	π	Pi; a constant (approx 3.1416), defined as the ratio of circumference of a circle to its diameter

5.4 ANALYTIC GEOMETRY

ISO Ref.	ITEM No.	SIGN or SYMBOL	DESCRIPTION
	5.4.1	x, y, z	Rectangular (Cartesian) coordinates of a point in space
	5.4.2	x, y	Rectangular coordinates of a point in a plane

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T21:08Z
Check the source to verify that this is the current version before use.

ISO Ref.	ITEM No.	SIGN or SYMBOL	DESCRIPTION
	5.4.3	α	Alpha; indicates direction angle with x-axis
	5.4.4	λ	Lambda; Indicates direction cosine (with x axis)
	5.4.4.1	l	Indicates corresponding direction number
	5.4.5	β	Beta; indicates direction angle with y-axis
	5.4.6	μ	Mu, Indicates direction cosine (with y-axis)
	5.4.6.1	m	Indicates corresponding direction number
	5.4.7	γ	Gamma; indicates direction angle with z-axis
	5.4.8	ν	Nu; Indicates direction cosine (with z-axis)
	5.4.8.1	n	Indicates corresponding direction number
	5.4.9	r, θ, ϕ	Spherical coordinates of a point in space
	5.4.10	r, θ	Polar coordinates of a point in a plane
	5.4.11	ψ	Psi; indicates angle from radius vector to tangent of plane curve
	5.4.12	r, θ, z	Cylindrical coordinates of a point in space
	5.4.13	p, s	Indicates intrinsic coordinates
	5.4.14	e	Eccentricity of a conic
	5.4.15	p	Semi-latus rectum
	5.4.16	m	Slope of a curve or line
	5.4.17	C	Circumference of a circle
	5.4.18	r	Radius of a circle
	5.4.19	D, d	Diameter
	5.4.20	p	Radius (of curvature)
	5.4.21	d	Perpendicular distance from a point to a line (length of normal)

5.5 TRIGONOMETRY

ISO Ref.	ITEM No.	SIGN or SYMBOL	DESCRIPTION
	5.5.1	$^{\circ}$ (superscript)	Indicates degree(s)
	5.5.2	θ	Angle
	5.5.3	$'$ (superscript)	Prime, indicates minutes
	5.5.4	$''$ (superscript)	Double prime; indicates seconds
ISO	5.5.5	sin	Sine of angle
ISO	5.5.6	cos	Cosine of angle
ISO	5.5.7	tan	Tangent of angle
ISO	5.5.8	cot	Cotangent of angle
ISO	5.5.9	sec	Secant of angle
	5.5.10	csc	Cosecant of angle
ISO	5.5.11	cosec	Cosecant of angle
	5.5.12	vers	Versed sine of angle. 1—cos θ
	5.5.13	covers	Coversed sine of angle
	5.5.14	hav	Haversine of angle. ½ (1—cos θ)
	5.5.15	cis θ	cos θ+i sin θ
ISO	5.5.16	arc sin or $\sin^{-1}$	Inverse sine (of); angle whose sine is
ISO	5.5.17	arc cos or $\cos^{-1}$	Inverse cosine (of); angle whose cosine is
ISO	5.5.18	arc tan or $\tan^{-1}$	Inverse tangent (of); angle whose tangent is
ISO	5.5.19	arc cot or $\cot^{-1}$	Inverse cotangent (of); angle whose cotangent is
ISO	5.5.20	arc sec or $\sec^{-1}$	Inverse secant (of); angle whose secant is
	5.5.21	arc csc or $\csc^{-1}$	Inverse cosecant (of); angle whose cosecant is
	5.5.22	$\sin^n x$	n^{th} power of sin x
	5.5.23	$\cos^n x$	n^{th} power of cos x

5.6 HYPERBOLIC FUNCTIONS

ISO Ref.	ITEM No.	SIGN or SYMBOL	DESCRIPTION
ISO	5.6.1	sinh	Hyperbolic sine
ISO	5.6.2	cosh	Hyperbolic cosine

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T21:08Z
Check the source to verify that this is the current version before use.

ISO Ref.	ITEM No.	SIGN or SYMBOL	DESCRIPTION
ISO	5.6.3	tanh	Hyperbolic tangent
ISO	5.6.4	coth	Hyperbolic cotangent
ISO	5.6.5	sech	Hyperbolic secant
	5.6.6	csch	Hyperbolic cosecant
	5.6.7	arc sinh or $\sinh^{-1}$	Inverse hyperbolic function (of); angle whose hyperbolic sine is
	5.6.8	arc cosh or $\cosh^{-1}$	Inverse hyperbolic function (of); angle whose hyperbolic cosine is
	5.6.9	arc tanh or $\tanh^{-1}$	Inverse hyperbolic function (of); angle whose hyperbolic tangent is
	5.6.10	arc coth or $\coth^{-1}$	Inverse hyperbolic function (of); angle whose hyperbolic cotangent is
	5.6.11	arc sech or sech^{-1}	Inverse hyperbolic function (of); angle whose hyperbolic secant is
	5.6.12	arc csch or csch^{-1}	Inverse hyperbolic function (of); angle whose hyperbolic cosecant is
	5.6.13	$\sinh^n x$	n^{th} power of sinh x
	5.6.14	$\cosh^n x$	n^{th} power of cosh x

5.7 CALCULUS

ISO Ref.	ITEM No.	SIGN or SYMBOL	DESCRIPTION
	5.7.1	d	Differential operator
	5.7.2	d^n	Differential operator of nth order
	5.7.3	$\frac{d}{dx}$	Derivative operator of first order with respect to x
	5.7.4	$\frac{d^n}{dx^n}$	Derivative operator of nth order with respect to x
	5.7.5	∂	Curly d; indicates partial differentiation
	5.7.6	D	Derivative operator
	5.7.7	D^n	Derivative operator of nth order
	5.7.8	$\dot{x}$, $\ddot{x}$	Indicates first, second, derivatives respectively, with respect to time (Newton's notation)
	5.7.9	$d^n y/dx^n$ or $d^n/dx^n y$	n^{th} order of the derivative of y with respect to x
	5.7.10	$'$ (superscript)	Prime; first derivative
	5.7.11	$''$ (superscript)	Double prime; second derivative
	5.7.12	$'''$ (superscript)	Triple prime; third derivative
	5.7.13	$\int$	Integral sign
	5.7.14	$\int_a^b$	Integral sign; indicating limits
	5.7.15	$\oint$	The line integral around a closed path
ISO	5.7.16	Δ	Delta; indicates increment
ISO	5.7.17	$\sum_i$	Sigma; indicates summation; sum of terms of index i
ISO	5.7.18	π_i	Pi; indicates product; product of terms of index i

5.8 SPECIAL FUNCTIONS

ISO Ref.	ITEM No.	SIGN or SYMBOL	DESCRIPTION
	5.8.1	$J_0(x)$, $J_1(x)$, $J_n(x)$	Bessel Functions, The notation recommended is G. N. Watson's Treatise, 1922, as endorsed by E. P. Adams in the Smithsonian Tables, 1922
	5.8.2	$B_0(x)$, $B_1(x)$, $B_2(x)$, ...	Bernoulli polynomials
	5.8.3	B_0, B_1, B_2, ...	Bernoulli numbers
	5.8.4	γ	Gamma; Euler's (Mascheroni's) constant. (0.5772 ...)
	5.8.5	$\Gamma(n)=\int_0^\infty x^{n-1}e^{-x}\,dx$	The Gamma function of the positive number n. Also called the factorial function

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T21:08Z
Check the source to verify that this is the current version before use.

ISO Ref.	ITEM No.	SIGN or SYMBOL	DESCRIPTION
	5.8.6	$B(m, n)=\int_0^1 x^{m-1}(1-x)^{n-1}\,dx$	The Beta function of any two positive numbers m and n
	5.8.7	$\Gamma(n, x)=\int_0^x e^{-t}\,t^{(n-1)}\,dt$	The incomplete Gamma function
	5.8.8	$B_x(m, n)=\int_0^x t^{m-1}(1-t)^{n-1}\,dt$	The incomplete Beta function

5.9 VECTOR ANALYSIS

ISO Ref.	ITEM No.	SIGN or SYMBOL	DESCRIPTION
	5.9.1	i, j, k	Orthogonal unit vectors
ISO	5.9.2	$A \cdot B$	Scalar product (dot product) of two vectors
ISO	5.9.3	$A \times B$	The vector product (cross product) of two vectors
	5.9.4	$\lvert A\rvert$	Absolute value of a vector, if $A=ai+bj$, then $\lvert A\rvert=(a^2+b^2)^{1/2}$
	5.9.5	$A\angle\theta$	Denotes the vector $A=ai+bj$, where $\theta=\arctan b/a$, $a=A\cos\theta$, $b=A\sin\theta$, $\lvert A\rvert = A$.
ISO	5.9.6	∇	Del; differential operator. $i\frac{\partial}{\partial x}+j\frac{\partial}{\partial y}+k\frac{\partial}{\partial z}$

6. NOTES

6.1 *International standardization agreements.* Certain provisions of this standard are the subject of International Standardization Agreement, ABC NAVY-STD-28A, Symbols and Abbreviations for Electrical and Electronics Diagrams, to which the U.S. Army subscribes and to which the Air Force interposes no objections. When reaffirmation, amendment, revision, or cancellation of this standard is proposed, which will effect or violate the international agreement concerned, the Preparing Activity will take appropriate reconcilliation action through international standardization channels, including departmental standardization offices, as required.

CUSTODIANS:
Army–EL
Navy–SH
Air Force–11

REVIEW Activities:
Army–EL, CE, WC, MU
Navy–SH, MC, EC, AS
Air Force–11

USER Activities:
Army–
Navy–OS, YD, CG
Air Force–

International Interest (see 6.1).

Preparing activity:
Navy–SH
(Project DRPR–0158)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T21:08Z
Check the source to verify that this is the current version before use.

APPENDIX

10.1 REVISED OR DELETED SYMBOLS

Symbols Formerly in MIL–STD–106			Recommended Symbols in MIL–STD–106A
2.21	$\equiv_x$	Indicates identity with all values of x for which both terms are defined.	5.2.22
2.24	$\not>$	Equal to or less than; not greater than.	5.2.27
2.29	N!	Factorial; continued product of all integral numbers from 1 to N, where N is an integral number.	5.2.33
2.30	^{n}(superscript numbers or letters)	Exponent; raised to the power of degree n (exponent indicates number of iterations).	5.2.34
2.31	$\sqrt[n]{\ }$	Radical sign; superscript n indicates index of degree of root. Index omitted in case of square root.	5.2.35
2.37	$a \cdot 10^n$	Scientific notation; notation by powers of 10.	5.2.41
2.45	$\log_a X$	Logarithm of X to base a.	5.2.49
2.46	$\log X$	Logarithm of X to base 10. (common system of logarithms).	5.2.50
2.47	$\ln X$	Logarithm of X to base e (Naperian system or Natural Logarithm).	5.2.51
2.51	$\lvert\ \rvert$	Vertical bars; indicates absolute value of the quantity inside the bars; vector magnitude; determinant.	5.2.55
2.52	$\lvert\ \rvert\quad\lvert\ \rvert$	Double vertical bars; indicates a matrix; set of quantities written in specific order of rows and columns.	5.2.56
3.14	$\overleftarrow{AB}$	Directed segment B to A.	5.3.14
4.4	l	Indicates directional cosine (with x axis).	5.4.4
4.6	m	Indicates directional cosine (y-axis).	5.4.6
4.8	n	Indicates directional cosine (with z-axis).	5.4.8
5.17	$[\sin f(x)]^n$	The nth power (of).	5.5.22 5.5.23
6.5	$[\sinh\ f(x)]^n$	nth power (of).	5.6.13
6.6	$[\cosh f(x)]^n$	nth power (of) etc.	5.6.14
7.1	d	Differential operator.	5.7.1
7.2	d^n	Differential operator of nth order.	5.7.2
7.3	$\frac{d}{dx}$	Derivitive operator of first order.	5.7.3
7.4	$\frac{d^n}{dx}$	Derivitive operator of nth order.	5.7.4
7.5	∂	Curly d; indicates partial differentiation.	5.7.5
7.6	D	Differential operator.	5.7.6
7.7	D^n	Differential operator of nth order.	5.7.7
7.9	$d^n y/dx^n$	Derivitive of nth order.	5.7.9
7.12	$\int, \iint, \iiint$	Integral signs.	5.7.13
7.13	$\int_i^m, \int_a^b, \int_a^b \int_c^d$	Integral signs; indicating index and limits.	5.7.14

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T21:08Z
Check the source to verify that this is the current version before use.

Symbols Formerly in MIL-STD-106			Recommended Symbols in MIL-STD-106A
8.1	$J_0(x)$, $J_1(x)$, $J_n(x)$.	Bessel Functions, The notation recommended is G. N. Watson's Treatise, 1922, as endorsed by E. P. Adams in the Smithsonia Tables, 1922.	5.8.1
8.2	$B_1, B_3, B_5, \ldots$ $B_1, B_2, B_3, \ldots$	Bernoulli numbers and polynomials.	5.8.2 5.8.3
8.4	$\Gamma(x) = {}_0\int^{\infty} x^{n-1} e^{-z} dx$	The Gamma function of the positive number n. Also called the factorial function.	5.8.5
8.5	$B(m, n) = {}_0\int^{1} x^{m-1} (1-x)^{n-1} dx$	The Beta function of any two positive numbers m and n.	5.8.6
8.6	$\Gamma x^{(n-1)} = {}_0\int^{xn}_{x} e^{-z} dx$	The incomplete Gamma function.	5.8.7
8.7	$B_z(m, n) = {}_0\int^{x} x^{m-1} (1-x)^{n-1} dx$	The incomplete Beta function.	5.8.8
9.1	$\bar{i}, \bar{j}, \bar{k}$	Vectors of unit magnitude.	5.9.1
9.2	$\bar{A} \cdot \bar{B}$	Scalar product (dot product) of two vectors.	5.9.2
9.3	$\bar{A} \times \bar{B}$	The vector product (cross product) of two vectors.	5.9.3
9.4	$A\angle\theta$	Indicates the vector $\bar{A} = a\bar{i} + b\bar{j}$ (or) $\bar{i}a + \bar{j}b$, where $a = \lvert\bar{A}\rvert \cos\theta$, $b = \lvert\bar{A}\rvert \sin\theta$ $\theta = \arctan b/a$, and $\lvert\bar{A}\rvert = (a^2 + b^2)^{1/2}$	5.9.5

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T21:08Z
Check the source to verify that this is the current version before use.

INDEX

A

$A\angle\theta$ 5.9.5
A • B 5.9.2
A X B 5.9.3
$a \cdot 10^n$ 5.2.41
a_{ij} 5.2.57
Absolute value 5.2.55
Add or substract 5.2.4
5.2.5
Addition 5.2.2
Alpha 5.4.3
Analytic geometry 5.4
Angle 5.3.1
5.5.2
between radius vector and tangent of curve 5.4.11
Approximate 5.2.20
Arc
cos 5.5.17
cosh 5.6.8
csc 5.5.21
csch 5.6.12
cot 5.5.19
coth 5.6.10
of a circle 5.3.15
sec 5.5.20
sech 5.6.11
sin 5.5.16
sinh 5.6.7
tan 5.5.18
tanh 5.6.9
Arrow 5.2.44
superscript 5.3.14
Asymptotic 5.2.21

B

Bars
vertical 5.2.55
5.9.4
vertical, double 5.2.56
Bernoulli
numbers 5.8.3
polynominals 5.8.2
Bessel functions 5.8.1
Beta 5.4.5
function 5.8.6
5.8.8
Braces 5.2.9
Brackets 5.2.8

C

Calculus 5.7
Chord of a circle 5.3.13
Circle 5.3.5
Circumference 5.4.17
Cis θ 5.5.15
Colon 5.2.15
Combinations 5.2.54
Congruent 5.3.9
Conics
diameter 5.4.19
eccentricity 5.4.14
radius of circle 5.4.18
radius (of curvature) 5.4.20
semi-latus rectum 5.4.15
slope of curve 5.4.16
Constant, Euler's 5.8.4
Coordinates
cartesian 5.4.1
cylindrical 5.4.12
intrinsic 5.4.13
polar 5.4.10
rectangular 5.4.1
5.4.2
spherical 5.4.9
Corresponds to 5.2.18
Cosecant 5.5.10
5.5.11
hyperbolic 5.6.6
Cosine 5.5.6
hyperbolic 5.6.2
Cotangent 5.5.8
hyperbolic 5.6.4
Cross product 5.9.3
Curly d 5.7.5
Cylindrical coordinates 5.4.12

D

Decimal point 5.2.42
Degrees 5.5.1
Del 5.9.6
Delta 5.7.16
Derivative 5.7.9
operator 5.7.3
5.7.4
Det 5.2.58
Determinant (vertical bars) 5.2.55
Diameter 5.4.19
Directed segment 5.3.14
Direction
angle 5.4.3
5.4.5
5.4.7
cosine 5.4.4
5.4.6
5.4.8
number 5.4.4.1
5.4.6.1
5.4.8.1
Division sign 5.2.14
Dot, centered (multiplication) 5.2.6
Dot product 5.9.2
Dots
(four) 5.2.16
(three) 5.2.48
5.3.12
(two) 5.2.15
Double prime 5.2.46

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T21:08Z
Check the source to verify that this is the current version before use.

E

e 5.2.52
Eccentricity of conic 5.4.14
Equal to or greater than 5.2.31
Equal to or less than 5.2.27
Equals by definition 5.2.23
Equiangular 5.3.11
Equivalent 5.2.17
Exclamation point 5.2.33
Exponent 5.2.34
5.2.37
Euler's constant 5.8.4

F

Four dots 5.2.16
Fractional index 5.2.36
Function
Bessel 5.8.1
exponential 5.2.38
5.2.39
hyperbolic 5.6
special 5.8

G

Gamma 5.4.7
5.8.4
function 5.8.5
5.8.7
Greater than, much greater than 5.2.28
5.2.29
5.2.30
Grouping 5.2.7
5.2.8
5.2.9
5.2.10
5.2.59

H

Half-space 5.2.59
Horizontal rule 5.2.13
Hyperbolic function 5.6

I

Identical to 5.2.22
Identify 5.2.22
Imaginery unit 5.2.40
Increment 5.7.16
Inequality 5.2.19
Infinity 5.2.43
Integral 5.7.13
5.7.14
5.7.15
Intrinsic coordinates 5.4.13
Inverse hyperbolic function 5.6.7
5.6.8
5.6.9
5.6.10
5.6.11
5.6.12

J

j
operator 5.2.40
orthogonal unit vector 5.9.1

L

l
direction number 5.4.4.1
Length of arc 5.3.15
Less than, much less than 5.2.24
5.2.25
5.2.26
Limits
arrows 5.2.44
summation 5.7.17
Log 5.2.49
5.2.50
5.2.51

M

m
direction number 5.4.6.1
slope 5.4.16
Mascheroni's constant 5.8.4
Matrix 5.2.56
Minus 5.2.3
Minus-plus 5.2.5
Multiplication 5.2.6
Mu 5.4.6

N

n
direction number 5.4.8.1
exponent 5.2.34
Negative value 5.2.3
Notation
Bernoulli numbers 5.8.2
Bernoulli polynominals 5.8.3
Bessel functions 5.8.1
Newton's 5.7.8
Scientific 5.2.41
variables or constants 5.2.45
5.2.46
5.2.47

O

Operator
derivative 5.7.3
5.7.4
5.7.6
5.7.7
differential 5.7.1
5.7.2
i 5.2.40
j 5.2.40
Overscript
vinculum 5.2.10

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T21:08Z
Check the source to verify that this is the current version before use.

P

p
  intrincis coordinate 5.4.13
  radius (of curvature) 5.4.20
  semi-latus rectum 5.4.15
Parallel 5.3.3
Parallelogram 5.3.6
Parenthesis 5.2.7
Percent 5.2.11
Permutation 5.2.53
Perpendicular 5.3.2
  distance 5.4.21
Pi 5.3.16
  5.7.18
Plus 5.2.2
Plus-minus 5.2.4
  5.2.5
Polar coordinates 5.4.10
Positive value 5.2.2
Power of
  cos s 5.5.23
  $\cosh^n x$ 5.6.14
  degree 5.2.36
  sin x 5.5.22
  $\sinh^n x$ 5.6.13
  superscript 5.2.34
  ten 5.2.41
Prime 5.2.45
  5.2.46
  5.2.47
  5.7.10
  5.7.11
  5.7.12
Product 5.7.18
  scalar 5.9.2
  vector 5.9.3
Psi 5.4.11

R

Radical 5.2.35
Radius
  of a circle 5.4.18
  of curvature 5.4.20
Radio 5.2.15
  of circumference 5.3.16
Root, extraction of 5.2.35

S

Secant of angle 5.5.9
Secant, Hyperbolic 5.6.5
Semi-latus rectum 5.4.15
Sigma 5.7.17
Signs
  absolute value 5.2.55
  addition 5.2.2
  arithmetic & algebra 5.2
  calculus 5.7
  geometry 5.3
  grouping 5.2.7
  5.2.8
  5.2.9
  5.2.10
  5.2.59
  hyperbolic functions 5.6
  limit 5.2.44
  minus 5.2.3
  plus 5.2.2
  similar 5.3.10
  subtraction 5.2.3
  trigonometry 5.5
  varies directly as 5.2.32
  vector analysis 5.9
Sigma 5.7.17
Sine
  coversed 5.5.13
  haversine 5.5.14
  hyperbolic 5.6.1
  of angle 5.5.5
  versed 5.5.12
Slope 5.4.16
Solidus 5.2.12
Space 5.2.59
Spherical coordinates 5.4.9
Square 5.3.7
  root 5.2.35
Subscript 5.2.60
Subtraction 5.2.3
Summation 5.7.17
Superscript
  degrees 5.5.1
  letters 5.2.34
  5.2.35
  5.2.36
  5.2.37
  numbers 5.2.34
  5.2.35
  5.2.36
  5.2.37
  primes 5.2.45
  5.2.46
  5.2.47
  5.5.3
  5.5.4
  5.7.10
  5.7.11
  5.7.12

T

Tangent 5.5.7
  5.6.3
Three dots 5.2.48
  5.3.12
Times 5.2.6
Trapezoid 5.3.8
Triangle 5.3.4
Triple prime 5.2.47
  5.7.12

U

Unit vector, orthogonal 5.9.1

V

Value
  absolute 5.2.55

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T21:08Z
Check the source to verify that this is the current version before use.

indicating 5.2.60
negative 5.2.3
positive 5.2.2
Variables 5.2.60
Vector
absolute value 5.9.4
angle 5.9.5
cross product 5.9.3
scalar product 5.9.2
unit, orthogonal 5.9.1
Vertical bars 5.2.55
5.2.56
Vinculum 5.2.10
5.3.13

X

x
direction angle 5.4.3
direction cosine 5.4.4
direction number 5.4.4.1
times (multiplication) 5.2.6
x, y rectangular coordinates 5.4.2
x, y, z Cartesian coordinates 5.4.1

Y

y
direction angle 5.4.5
direction cosine 5.4.6
direction number 5.4.6.1

Z

z
direction angle 5.4.7
direction cosine 5.4.8
direction number 5.4.8.1

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T21:08Z
Check the source to verify that this is the current version before use.